

MATH3026/4022 Lecture 9

Model Diagnostics

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Outline

- ▶ Steps for basic model checking
- ▶ Hypothesis testing
- ▶ Model residuals
- ▶ Goodness of fit tests
- ▶ Model selection using information criteria

Introduction

During previous lectures, we have seen how time series models may be fitted and parameters of interest estimated.

Once we've fitted a model, a natural question would be: 'How do we check whether or not a model is a good fit to our data?'

In this lecture, we'll explore how a fitted time series model may be assessed to see how well it fits a given dataset.

Suppose that we have fitted a model to some sampled time series data and obtained parameter estimates (say $\theta_1, \dots, \theta_q$ and/or $\phi_1, \dots, \phi_p$).

Typically, our first check would be

1. **Are the parameter estimates in the admissible regions for stationarity and/or invertibility?**

Thankfully, if we use the `arima` command in R this check is done for us when fitting models and R will only return admissible parameter estimates for a given model type.

2. Can we perform hypothesis tests to identify the most appropriate model?

When fitting models to data using linear regression, we recall that a **parsimonious model** is desirable.

That is, a model that fits the data well but with as small a number of parameters as possible (for a good fit).

Fitting a model to time series data is no different. We would like a model that fits well but with a reasonable number of parameters.

For example, we might consider whether or not it is necessary to fit an $AR(2)$ model when we could fit an $AR(1)$ model instead.

Model Diagnostics

The $AR(p)$ model has one extra parameter (ϕ_p) than the $AR(p-1)$ model.

To decide whether or not an $AR(p)$ model is preferable to an $AR(p-1)$ model, we might like to test the hypotheses

$$H_0: \phi_p = 0 \text{ versus } H_1: \phi_p \neq 0.$$

Where

$$X_t = \phi_1 X_{t-1} + \dots + \phi_{p-1} X_{t-(p-1)} + \phi_p X_{t-p} + Z_t.$$

If $\hat{\phi}_p$ is the **maximum likelihood estimator** of ϕ_p then we know that, asymptotically:

$$\hat{\phi}_p \sim \mathcal{N}(\phi_p, \text{Var}(\hat{\phi}_p))$$

Then, assuming that H_0 is true (i.e. $\phi_p = 0$)

$$\begin{aligned}\hat{\phi}_p \sim \mathcal{N}\left(0, \text{Var}\left(\hat{\phi}_p\right)\right) &\implies \frac{\hat{\phi}_p}{\sqrt{\text{Var}\left(\hat{\phi}_p\right)}} \sim \mathcal{N}(0, 1) \\ &\implies \frac{\hat{\phi}_p}{s.e.(\hat{\phi}_p)} \sim \mathcal{N}(0, 1).\end{aligned}$$

where $s.e.(\hat{\phi}_p)$ is the standard error of $\hat{\phi}_p$.

Model Diagnostics

We see that the test statistic

$$\frac{\hat{\phi}_p}{s.e.(\hat{\phi}_p)}$$

has, asymptotically, a standard normal distribution.

As a basic rule of thumb, we would reject H_0 if

$$\left| \frac{\hat{\phi}_p}{s.e.(\hat{\phi}_p)} \right| > 2.$$

Example 9.1

A sample of time series data of size $n = 1000$ was collected. It is thought that the data are from an $AR(p)$ process and both an $AR(1)$ model and an $AR(2)$ model were fitted to the data, with model coefficients and standard error estimates presented below.

$AR(1)$ model:

Coefficients:

	ar1	intercept
	0.4854	0.0459
s.e.	0.0276	0.0614

$AR(2)$ model:

Coefficients:

	ar1	ar2	intercept
	0.4623	0.0477	0.0462
s.e.	0.0316	0.0316	0.0644

Example 9.1

Which model should be fitted to the observed data? Justify your answer.

Model Diagnostics

If H_0 is rejected then we should prefer to fit an $AR(p)$ model over an $AR(p-1)$ model.

This approach is not only valid for AR models, indeed it holds for linear model parameters (i.e. ϕ_i and θ_j terms) for $ARMA(p, q)$ models.

Remember that AR and MA models are simply special cases of $ARMA$ models.

To perform the hypothesis test, we require the standard error of the estimated parameter.

This is not necessarily straightforward, especially when a numerical method has been used to obtain the maximum likelihood estimates.

Model Diagnostics – Standard Error Estimation

Suppose that our parameters of interest when fitting a time series model are

$$\boldsymbol{\theta} = (\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, \mu, \sigma_z^2)^\top.$$

Now let $\ell(\boldsymbol{\theta}; \mathbf{x})$ denote the model's log-likelihood function, given a sample of data $\mathbf{x}$

We assume that a numerical optimisation method has been used to maximise $\ell(\boldsymbol{\theta}; \mathbf{x})$ and obtain the maximum likelihood estimates $\hat{\boldsymbol{\theta}}$.

Model Diagnostics – Standard Error Estimation

The **observed information matrix** (with respect to $\ell = \ell(\boldsymbol{\theta}; \mathbf{x})$) is defined

$$\mathcal{J}(\hat{\boldsymbol{\theta}}; \mathbf{x}) = \left[\begin{array}{cccc} -\frac{\partial^2 \ell}{\partial \phi_1^2} & -\frac{\partial^2 \ell}{\partial \phi_1 \partial \phi_2} & \cdots & -\frac{\partial^2 \ell}{\partial \phi_1 \partial (\sigma_z^2)} \\ -\frac{\partial^2 \ell}{\partial \phi_2 \partial \phi_1} & -\frac{\partial^2 \ell}{\partial \phi_2^2} & \cdots & -\frac{\partial^2 \ell}{\partial \phi_2 \partial (\sigma_z^2)} \\ \vdots & \vdots & \vdots & \vdots \\ -\frac{\partial^2 \ell}{\partial (\sigma_z^2) \partial \phi_1} & -\frac{\partial^2 \ell}{\partial (\sigma_z^2) \partial \phi_2} & \cdots & -\frac{\partial^2 \ell}{\partial (\sigma_z^2)^2} \end{array} \right] \bigg|_{\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}}.$$

When the sample n is large and, subject to regularity conditions, the observed information matrix may be used to approximate the Fisher information matrix $\mathcal{I}(\boldsymbol{\theta})$.

Model Diagnostics – Standard Error Estimation

Recall that the Fisher information matrix is defined

$$\mathcal{I}(\boldsymbol{\theta}) = \begin{bmatrix} \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial \phi_1^2} \right) & \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial \phi_1 \partial \phi_2} \right) & \cdots & \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial \phi_1 \partial (\sigma_z^2)} \right) \\ \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial \phi_2 \partial \phi_1} \right) & \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial \phi_2^2} \right) & \cdots & \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial \phi_2 \partial (\sigma_z^2)} \right) \\ \vdots & \vdots & \vdots & \vdots \\ \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial (\sigma_z^2) \partial \phi_1} \right) & \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial (\sigma_z^2) \partial \phi_2} \right) & \cdots & \mathbb{E} \left(-\frac{\partial^2 \ell}{\partial (\sigma_z^2)^2} \right) \end{bmatrix}.$$

Asymptotically, (i.e. as $n \rightarrow \infty$) the distribution of the maximum likelihood estimators of $\boldsymbol{\theta}$ is

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}_k(\boldsymbol{\theta}, \mathcal{I}(\boldsymbol{\theta})^{-1}).$$

where k is the dimension of the vector $\boldsymbol{\theta}$.

Model Diagnostics – Standard Error Estimation

Using $\mathcal{J}(\hat{\boldsymbol{\theta}}; \mathbf{x})$ as an approximation of $\mathcal{I}(\boldsymbol{\theta})$, we deduce that the variance-covariance matrix of $\hat{\boldsymbol{\theta}}$ may be estimated as

$$\text{Var}(\hat{\boldsymbol{\theta}}) \approx \mathcal{J}(\hat{\boldsymbol{\theta}}; \mathbf{x})^{-1}.$$

Standard error estimates of $\hat{\boldsymbol{\theta}}$ can be computed using the diagonal elements of $\mathcal{J}(\hat{\boldsymbol{\theta}}; \mathbf{x})^{-1}$.

In R, when using numerical optimisation, the `hessian` matrix of a fitted model may be inverted (assuming a likelihood-based fit) and used as an approximation of $\mathcal{J}(\hat{\boldsymbol{\theta}}; \mathbf{x})^{-1}$.

3. Can we check how well the chosen model fits the data?

One way of doing this is to **simulate** data from the fitted model.

Then, we take the set of simulated data and produce sample ACFs, PACFs etc.

These can then be compared with those created using the observed set of data and checked to see how similar they are.

If the model fits the data well then the plots should look very similar.

Note that we can also compare theoretical ACFs, PACFs etc. with sample ACFs, PACFs from the observed data.

4. Check that the model does not overfit.

As mentioned in point 2., we should seek a **parsimonious** model – that is a simple model with as few parameters as possible that fits the data well.

For example, if we think that an $ARMA(2,2)$ model is appropriate we should also try fitting $ARMA(2,1)$, $ARMA(1,2)$ and $ARMA(1,1)$ models and compare the fits using ideas in points 1.-3. before deciding on a final model.

Finally, if the length of time over which the time series occurs is very long, then it may be sensible to partition the time series and fit separate models to each partition to ensure that the model fits well over a long period of time.

Model Diagnostics: Residuals

When fitting linear regression models, we know that model residuals are often examined when checking model fit.

Residuals are also used when checking the fit of time series models.

First we will define residuals for the $ARMA(p, q)$ model.

Suppose that we have an $ARMA(p, q)$ model, written

$$\phi(B) (X_t - \mu) = \theta(B) Z_t$$

and we denote $\hat{\mu}, \hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q, \hat{\sigma}_z^2$ to be the estimated model parameters. We define

$$\begin{aligned}\hat{\phi}(B) &= 1 - \hat{\phi}_1 B - \dots - \hat{\phi}_p B^p; \\ \hat{\theta}(B) &= 1 + \hat{\theta}_1 B + \dots + \hat{\theta}_q B^q.\end{aligned}$$

Model Diagnostics: Residuals

The equation of the fitted model is then

$$\hat{\phi}(B)(X_t - \hat{\mu}) = \hat{\theta}(B)Z_t.$$

To estimate the residuals, we re-arrange the above equation to solve for Z_t , yielding

$$\hat{Z}_t = [\hat{\theta}(B)]^{-1}\hat{\phi}(B)(X_t - \hat{\mu}).$$

Sometimes the residuals, $\{\hat{Z}_t\}$, may be known as the 'white noise estimates'.

In addition, solving the above equation is always possible when the time series model is invertible.

Model Diagnostics: Residuals

If a model is a good fit to the observed data then the residuals should look like observations from a white noise process.

Indeed, if $\mu, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, \sigma_z^2$ are known then it can be shown that

$$r_k(\hat{Z}_t) \sim \mathcal{N}\left(0, \frac{1}{n}\right)$$

where $r_k(\hat{Z}_t)$ is the autocorrelation of the residuals $\{\hat{Z}_t\}$ at lag k and $r_j(\hat{Z}_t)$ and $r_k(\hat{Z}_t)$ are independent where $j \neq k$.

Typically, $\mu, \phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q, \sigma_z^2$ are unknown but, nonetheless, we assume that, approximately

$$r_k(\hat{Z}_t) \sim \mathcal{N}\left(0, \frac{1}{n}\right) \text{ independently.}$$

Model Diagnostics: Residuals

If the residuals are white noise then we might expect the ACFs $r_k(\hat{Z}_t)$ to lie within $\pm$ two standard deviations of the expected value of the ACF (i.e. 0).

Hence, if we plot the sample ACFs of the residuals against the lag then we'd expect each of these ACFs to lie in the interval $\left(-\frac{2}{\sqrt{n}}, \frac{2}{\sqrt{n}}\right)$.

In fact, we have already considered plots of ACFs to identify time series models in Lecture 7.

Here we follow a similar approach with the model residuals to ensure that these look like white noise.

Portmanteau test for lack of fit

Suppose that we have observed n data points and we have fitted an $ARMA(p, q)$ model.

We introduce the statistic Q_k where

$$Q_k = n \sum_{h=1}^k (r_h(\hat{Z}_t))^2 = n [r_1^2 + \dots + r_k^2]$$

where, for simplicity $r_h(\hat{Z}_t) = r_h$.

Note that

$$\sqrt{n}r_h \sim \mathcal{N}(0, 1) \implies nr_h^2 \sim \chi_1^2.$$

It can be shown that

$$Q_k \sim \chi_{k-p-q}^2$$

if the residuals are from a white noise process.

Portmanteau test for lack of fit

If the model fits the data well (i.e. if the residuals are white noise) then we'd expect an observed value of the test statistic Q_k , based on our data, to be a likely observation from a χ^2_{k-p-q} distribution.

Hence, if we calculate Q_k and our value does not appear to be a draw from a χ^2_{k-p-q} distribution (i.e. is an extreme value in the tail of the distribution's pdf) then we might conclude that the model does not fit the data well.

However, how do we choose k ?

We wouldn't want to do multiple tests for different values of k because this could lead to problems encountered when multiple testing (i.e. false positives).

In fact, this test is only accurate where k is fairly large (usually we might choose $k \geq 20 + p + q$) and, as such, the test is often avoided in favour of a Ljung-Box test (which we'll now discuss).

Ljung-Box test

The Ljung-Box test is an improvement on the portmanteau test from the previous slide and is used to test the hypotheses

H_0 : The residuals are independent (i.e. ACFs are zero);

H_1 : The residuals are not independent (i.e. ACFs are non-zero).

The Ljung-Box test statistic is

$$Q_k = n(n+2) \sum_{h=1}^k \frac{r_h^2}{n-h}.$$

Under H_0 (i.e. assuming H_0 is true) asymptotically,

$$Q_k \sim \chi_{k-p-q}^2.$$

Note that the model is likely to be a good fit if H_0 is **retained** (i.e. where H_0 is not rejected).

Model Residuals - Diagnostics

Having fitted a particular model, we should extract the model residuals and produce the following plots:

1. **A time plot of the residuals** (i.e. a plot of $\hat{Z}_t$ versus t). We should examine this to determine whether or not the time series for the residuals looks like white noise.
2. **A plot of the sample ACFs for the residuals.** The residuals should be uncorrelated so we'd expect see small values of r_h for $h \geq 1$ (typically in the interval $(-2/\sqrt{n}, 2/\sqrt{n})$).
3. **A plot of the P-values of Ljung-Box test for different lags.** If the model fits the data well then we should expect these P-values to be non-significant (typically > 0.05).

We'll explore such plots using an example of a model fitted in R.

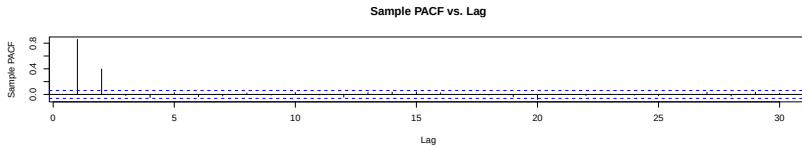
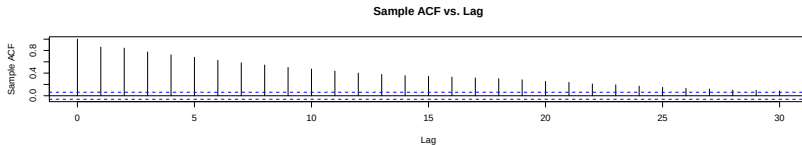
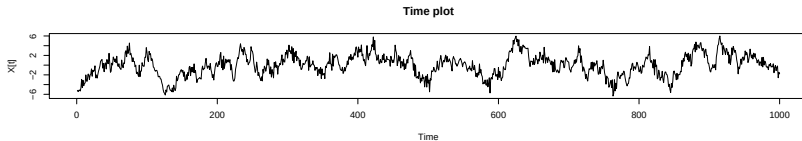
Model Diagnostics - An Example

We simulate 1000 observations from an $AR(2)$ process with $\mu = 0$, $\phi_1 = 0.5$, $\phi_2 = 0.4$ and $\sigma_z^2 = 1$

The R command used to simulate the data is

```
x<-arima.sim(n=1000,model=list(ar=c(0.5,0.4),sd=1))
```

Model Diagnostics - An Example



Model Diagnostics - An Example

Now we fit an $AR(2)$ model (the correct model) to these simulated data and examine some diagnostic plots.

The R code used to fit the model is

```
model.ar2<-arima(x,order=c(2,0,0),method="ML")
```

The R output is

Call:

```
arima(x = x, order = c(2, 0, 0))
```

Coefficients:

	ar1	ar2	intercept
	0.4442	0.4645	-0.0551
s.e.	0.0280	0.0280	0.3299

```
sigma^2 estimated as 0.9331:  log likelihood = -1385.14,  
aic = 2778.28
```

Model Diagnostics - An Example

We see that the model parameter estimates lie close to the true parameter values (as we should expect).

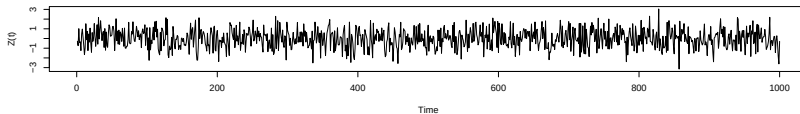
To extract the model residuals, we use the command

```
Z_hat<-residuals(model.ar2)
```

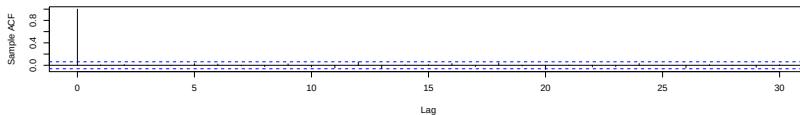
Next we'll examine plots of the estimated residuals together with the sample ACF and P-values for the Ljung-Box statistic at different lags.

Model Diagnostics - An Example

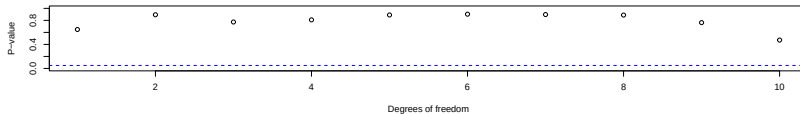
Time plot of residuals



Residuals: Sample ACF vs. Lag



Ljung-Box test P-values



Model Diagnostics - An Example

These plots look reasonable and indicate that the model appears to fit the data. That is

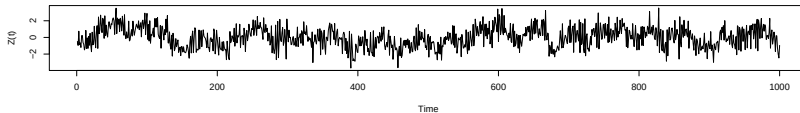
- ▶ The estimated residuals look like white noise in the time plot.
- ▶ The estimated residuals appear to be uncorrelated (from the ACF plot and the P-values for the Ljung-Box test statistics).

Now suppose we fit an incorrect model to the data.

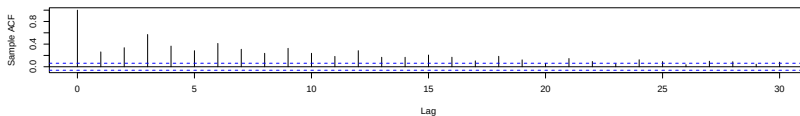
The next slide shows similar plots for the residuals, but where we've fitted an $MA(2)$ model instead.

Model Diagnostics - An Example

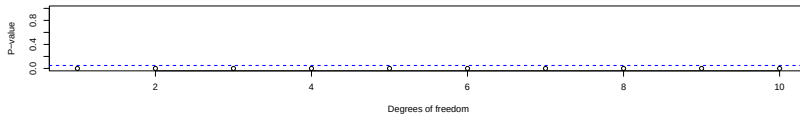
Time plot of residuals



Residuals: Sample ACF vs. Lag



Ljung-Box test P-values



Model Diagnostics - An Example

Examining these plots, it is clear that the residuals do not look like white noise.

Naturally, we should expect this since we have fitted an $MA(2)$ model to data that were simulated from an $AR(2)$ process!

Residuals - Modifying a Model

Having examined diagnostic plots of the residuals, suppose that the residuals don't look like white noise, how should we modify a model?

Suppose that the correct (but unknown) model is

$$\phi(B)X_t = \theta(B)Z_t$$

but that, instead, an incorrect model written

$$\phi_0(B)X_t = \theta_0(B)Z'_t$$

is used.

Residuals - Modifying a Model

We can see easily that the residuals from the incorrect model (i.e. the Z'_t terms) are correlated since

$$\begin{aligned}Z'_t &= \theta_0^{-1}(B)\phi_0(B)X_t \\ &= \theta_0^{-1}(B)\phi_0(B)\phi^{-1}(B)\theta(B)Z_t\end{aligned}$$

whereupon

$$\theta_0(B)\phi(B)Z'_t = \phi_0(B)\theta(B)Z_t.$$

The above equation indicates that the residuals Z'_t are an $ARMA(p', q')$ process for some p', q' (so clearly not white noise).

Example 9.2

Example 9.3

Example 9.4

Model Selection

When selecting a model we are interested in choosing the most appropriate model from a variety of competing possible models.

As noted previously, a key aim is to select a **parsimonious** model - that is a model that fits the data well but that does not contain too many parameters.

If we have used a maximum likelihood method to fit a model, then we can use an **information criterion** to select a model.

Akaike Information Criterion (AIC)

The Akaike information criterion (AIC) is defined

$$\text{AIC} = -2 \log(\text{Maximised Likelihood}) + 2k$$

Here, the 'maximised likelihood' is the likelihood function evaluated at the maximum likelihood estimates and k is the number of model parameters (i.e. the number of parameters that we estimated when maximising the likelihood function).

Once again, note that $\log =$ natural logarithm.

If we use the AIC to select an $ARMA(p, q)$ model then the chosen model is that which has the smallest AIC value.

Bayesian Information Criterion (BIC)

The Bayesian information criterion (BIC) is defined

$$\text{BIC} = -2 \log(\text{Maximised Likelihood}) + (\log n)k$$

where n is the sample size and k is the number of model parameters.

The BIC includes a stronger penalty on the number of parameters when compared to the AIC and typically results in a smaller number of parameters in the chosen model than when a model is chosen using the AIC.

Where the sample size is small, we should use the **corrected AIC** in place of the AIC.

The corrected AIC (written AIC_c) is defined

$$AIC_c = AIC + \frac{2k(k+1)}{n-k-1}.$$

Lecture 9: Learning Outcomes

- ▶ Understand how hypothesis tests can be used and constructed to compare nested time series models.
- ▶ Be aware of the importance of selecting a **parsimonious model** when fitting models to data.
- ▶ Know how the **residuals** are defined for a fitted time series model and understand how a plot of the ACF of the residuals may be interpreted with regard to checking model fit.
- ▶ Understand how the portmanteau test and Ljung-Box test may be used to assess model fit.
- ▶ Know the definitions of the AIC, BIC and AIC_c and understand how these criteria may be used for model selection.